GrowthLab decision memo

Product growth & experimentation analytics · public data (GA4 sample, Cookie Cats, Hillstrom) · all numbers computed in this repo

Key findings

1. The biggest loss is before any product is seen. Of 238,621 new GA4 visitors, only 21.0% view a product within 7 days (95% CI 20.8%–21.2%); 0.75% buy. No channel, device or country differs from the rest after Holm, and the device gap is +0.03 pp raw vs +0.03 pp within-channel (no Simpson effect).

2. Few visitors return: D7 return 7.9% (CI 7.7%–8.0%); early-November cohorts retained best, December holiday cohorts worst (cohorts differ at every age, Holm). First-day page depth is the strongest positive association with 30-day return (OR 1.62 per SD, CI 1.57–1.68): an association, likely partly intent.

3. Experiments: moving the Cookie Cats gate to level 40 LOWERED D7 retention by 0.82 pp (CI -1.33 to -0.31). Both Hillstrom e-mails raised conversion (Mens +0.68 pp, Womens +0.31 pp, Holm); of 44 subgroups examined 17 survive Holm, and only Womens × newbie shows a real interaction.

Recommendations

Cookie Cats - DO NOT SHIP gate_40 (keep level 30). D7 CI excludes any gain; D1 guardrail does not pass non-inferiority (lower bound -1.24 pp vs -1.0 pp margin); rounds stable. Revenue, the real guardrail, is missing. Next: test an earlier gate or shorter wait with revenue/player pre-specified.

Hillstrom - SHIP the Mens e-mail as default (larger conversion, visit and spend lift); target the Womens e-mail at new customers, the only Holm-supported interaction. Uplift targeting is exploratory and, at the assumed $0.10 per e-mail and $1 per visit, loses money either way: a scenario, not a plan.

Funnel - the sizing scenario (1 pp less first-visit drop-off) is worth about $5,888 over 12 weeks of this obfuscated sample: small and uncertain. Investigate landing-page relevance first.

First test to run

Randomize new visitors 50/50 to a 'continue browsing' landing module vs control. Primary: return in [t0+24h, t0+31d) (baseline 8.1%). Manipulation check: first-day page views. Guardrail: 7-day purchase conversion (non-inferiority). MDE 1.0 pp needs 12,387 per arm (about 9 days of traffic + 31-day window); 0.5 pp needs 48,272 per arm (about 66 days total).

What the data cannot tell us

Revenue impact of the gate change; D30+ and long-term effects; novelty effects (no timestamps); true sign-up or account behaviour (GA4 has cookies and first_visit only, obfuscated); whether driver associations are causal. Needed: revenue per player, 30-90 day windows, real sign-up/account data, and randomized tests of the drivers.